NOZZLE DESIGN USING METHOD OF CHARACTERISTICS

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22B0054
8th November 2024

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1 Introduction to the Method of Characteristics

The Method of Characteristics (MoC) is essential for analyzing nonlinear partial differential equations (PDEs) that arise in supersonic and compressible fluid flows. The velocity potential equation for compressible flows is inherently hyperbolic in supersonic regimes, which allows the use of characteristic lines to reduce the governing PDE into a set of ordinary differential equations (ODEs).

In particular, the MoC simplifies the analysis of the velocity potential equation:

$$\left(1-\frac{u^2}{a^2}\right)\frac{\partial^2\phi}{\partial x^2}+\left(1-\frac{v^2}{a^2}\right)\frac{\partial^2\phi}{\partial y^2}-\frac{2uv}{a^2}\frac{\partial^2\phi}{\partial x\partial y}=0, \tag{1}$$

where ϕ is the velocity potential, u and v are the velocity components in the x and y directions, respectively, and a is the speed of sound.

1.1 Characteristic Lines and Compatibility Relations

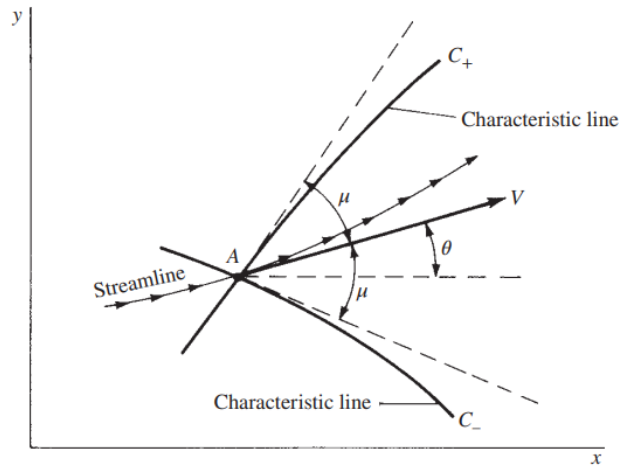


Figure 1: Left and right running characteristic lines through a point [2]

For two-dimensional, steady, inviscid, and irrotational supersonic flow, characteristic lines are defined by the slopes:

$$\frac{dy}{dx}=\tan(\theta\pm\mu), \tag{2}$$

where θ is the flow angle, and $\mu=\arcsin(1/M)$ is the Mach angle, with M representing the Mach number. These characteristic lines, denoted as C^+ and C^-, correspond to left- and right-running characteristic directions in the flow field.

The compatibility relations along these characteristics can be defined as:

$$d\theta=\pm\frac{\sqrt{M^2-1}}{M}\frac{dV}{V}, \tag{3}$$

where V is the flow speed. Integration of these relations yields important flow variables and allows us to construct a characteristic mesh, where flow properties such as M, θ, and pressure can be calculated iteratively.

2 Mathematical Formulation of the Velocity Potential Equation

The velocity potential ϕ is governed by:

$$\frac{\partial}{\partial x}\left[\left(1-\frac{u^2}{a^2}\right)\frac{\partial\phi}{\partial x}\right]+\frac{\partial}{\partial y}\left[\left(1-\frac{v^2}{a^2}\right)\frac{\partial\phi}{\partial y}\right]-\frac{\partial}{\partial x}\left(\frac{uv}{a^2}\frac{\partial\phi}{\partial y}\right)=0. \tag{4}$$

Rewriting in terms of characteristic slopes, we find the governing characteristic equations become:

$$\frac{dy}{dx} = -\frac{uv/a^2}{1-u^2/a^2} \pm \frac{\sqrt{(u^2+v^2)/a^2-1}}{1-u^2/a^2}. \tag{5}$$

Substituting $u = V\cos\theta$ and $v = V\sin\theta$, which simplifies the equation to

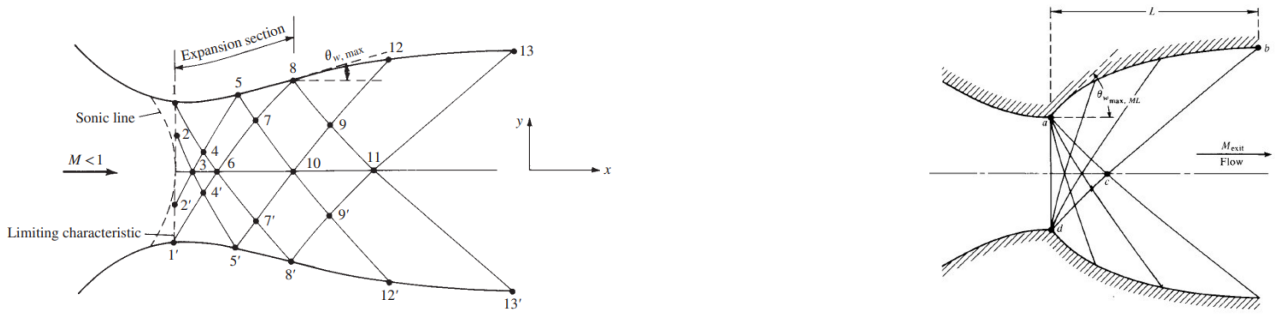
$$\left(\frac{dy}{dx}\right)_{char} = \tan(\theta \mp \mu)$$

3 Design of a Supersonic Nozzle Using MoC

The design of a supersonic nozzle typically involves a *convergent-divergent* configuration. The purpose of the divergent section is to accelerate the flow beyond Mach 1, facilitating a smooth transition from subsonic to supersonic flow. MoC is particularly useful in designing the nozzle's divergent section to achieve an optimal contour that avoids shocks.

3.1 Setting Up the Characteristics Mesh

To design the supersonic portion of the nozzle, a mesh of characteristic lines is generated. An initial data line is established, downstream from which characteristic lines (i.e., C^+ and C^- characteristics) are calculated. Key points of interest, such as internal points and wall points, are determined by the intersection of these characteristic lines.



(a) General 2D Nozzle[2]

(b) Minimum Length 2D Nozzle[3]

Figure 2: Characteristic Mesh

3.2 Equations for Internal and Wall Points

3.2.1 Internal Points

For internal points, two intersecting characteristics are used to solve for unknowns. Given points 1 and 2, point 3 is determined by the intersection of the C^- characteristic from point 1 and the C^+ characteristic from point 2. Compatibility relations yield:

$$\theta_3 + \nu(M_3) = (K^-)_{\text{characteristic from 1}}, \tag{6}$$

$$\theta_3 - \nu(M_3) = (K^+)_{\text{characteristic from 2}}, \tag{7}$$

where K^- and K^+ are constants along the C^- and C^+ characteristics, respectively.

3.2.2 Wall Points

For wall points, a C^- characteristic intersects the wall at a known angle. The wall angle θ_w and Prandtl-Meyer function ν at the wall point provide the following relation:

$$\theta_w + \nu(M) = K^-, \tag{8}$$

allowing all other flow properties at the wall to be determined.

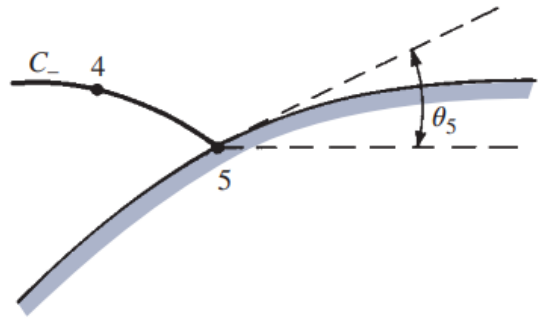


Figure 3: Wall Point [2]

3.3 Nozzle Contour Calculation

To achieve a specified exit Mach number M_e, the wall contour is calculated using the characteristic mesh. For instance, if we want to maintain shock-free, isentropic flow, the angle θ changes in the expansion and straightening sections:

$$\theta_{\max} = \nu(M_e) - \nu(1), \tag{9}$$

where $\theta_{\max}$ denotes the maximum wall angle in the expansion section. This angle decreases to zero in the straightening section, allowing a smooth exit.

4 Designing a Minimum-Length Nozzle[1]

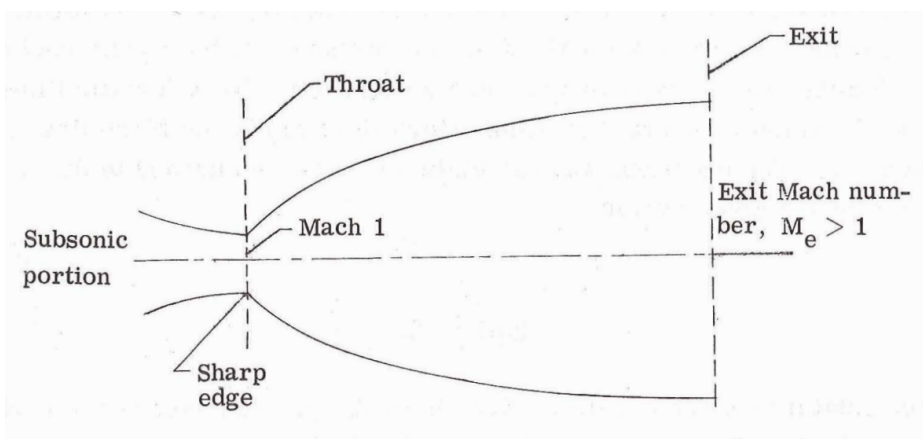


Figure 4: A Sharp-Edge or Minimum Length Nozzle [1]

In standard nozzle designs, a limiting characteristic line must be defined between the subsonic and supersonic regions at the throat, marking the boundary where the expansion begins. Determining this limiting characteristic requires detailed data on the flow conditions near the throat, adding complexity to the design. However, with a minimum-length nozzle, we avoid this complexity by starting the characteristic lines directly at the throat, allowing the supersonic flow to expand smoothly through the nozzle.
To design a minimum-length nozzle using MoC, we make the following assumptions:

- The flow is steady, inviscid, and two-dimensional.
- The nozzle operates under isentropic conditions, ensuring smooth, shock-free expansion.
- The flow reaches Mach 1 at the throat, marking the transition from subsonic to supersonic flow.

The process of designing a minimum-length nozzle using MoC is as follows:

- **Define Initial Conditions at the Throat:** With the flow at Mach 1 at the throat, we use this point as the starting line for generating the characteristic lines, eliminating the need for a limiting characteristic. The initial Mach angle at the throat is 90°, setting up the conditions for expansion.
- **Generate Characteristic Lines from the Throat:** By initiating C^+ and C^- characteristic lines directly from the throat, we create a straightforward characteristic mesh in the divergent section. Each intersection within this mesh provides a grid point where flow properties like Mach number, pressure, and flow angle are calculated.
- **Construct the Nozzle Wall Contour:** The nozzle wall is defined by the points where the C^+ characteristic lines intersect the wall, dictating the required wall angle for achieving the target exit Mach number. The expansion continues until the wall angle decreases to zero at the nozzle exit, achieving the desired supersonic flow profile over the shortest possible length.
- **Calculate Flow Properties at the Exit:** At the nozzle exit, we determine the final Mach number, pressure, and temperature, ensuring the nozzle meets specified supersonic conditions.

5 How to run the code

The code for the Method of Characteristics is written in the form of a Jupyter Notebook with Python 3 (ipykernel) as the kernel. NumPy, math and SciPy are the three packages used for the calculations done in the code and the plots are made using matplotlib. Although the code can run on a local machine, to avoid the issue of missing packages, Please run the code using the following LINK. Make sure the cells are executed in the order to avoid errors.

6 Final Nozzle Design

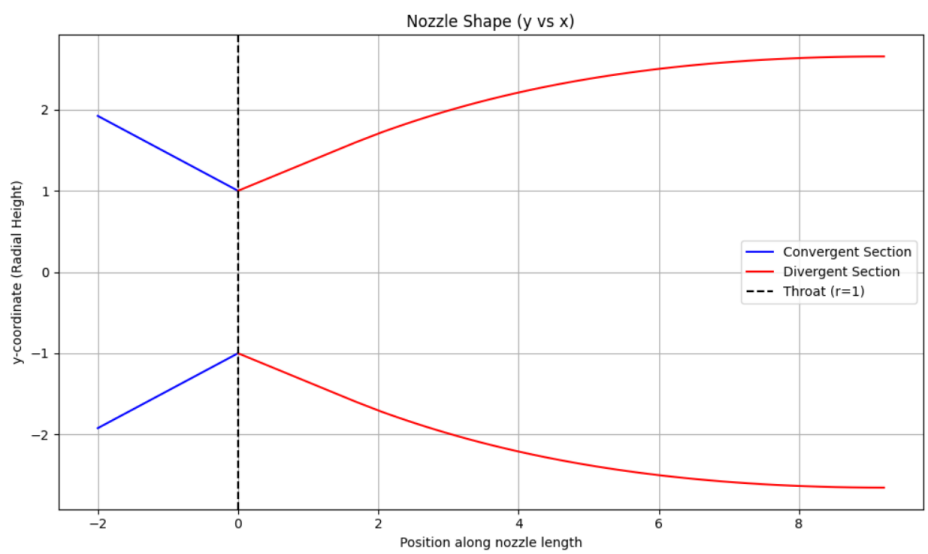


Figure 5: Designed Nozzle Contour

The inlet Mach number was established at 0.32 which is in line with standard rocket inlet conditions. During rocket nozzle design the features of the converging section were the smoothness of the contour as well as compliance with the area-Mach number relation. These criteria were systematically set forth in the design of the nozzle. In the case of the diverging section, the Method of Characteristics was used, which allowed for the accurate specification of the expansion contour, the characteristic plot of which can be seen from the code.

7 Simulation for the nozzle

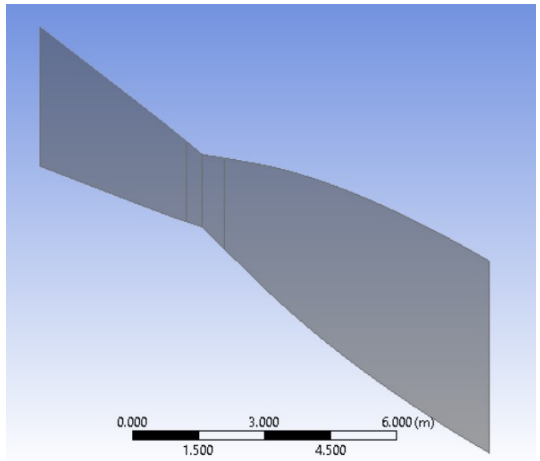


Figure 6: Surface CAD of the Nozzle

To verify that the output Mach number reaches 2.5, we conducted an inviscid 2D flow simulation in ANSYS Fluent[4]. The nozzle CAD model was created in SolidWorks, and the simulation yielded an output Mach number of 2.47, which closely approximates the target of 2.5. The minor discrepancy may be attributed to meshing quality and sharp edges in the simulation geometry.

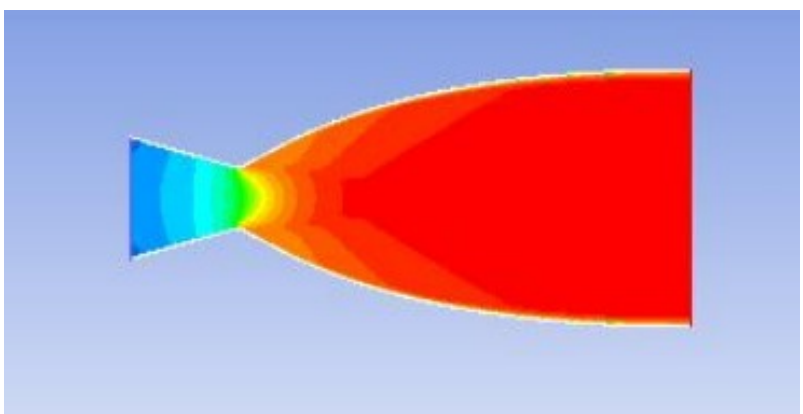


Figure 7: Mach Number Contour after the simulation

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[1] M. R. Vanco and L. J. Goldman, "Computer program for design of two-dimensional supersonic nozzle with sharp-edged throat," Lewis Research Center, Cleveland, Ohio, Tech. Rep., 1965, nASA Technical Report. [Online]. Available: https://ntrs.nasa.gov/api/citations/19650018105/downloads/19650018105.pdf

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[3] M. KHAN, S. Kumar, M. Sharath, and H. Chowdary, "Design of a supersonic nozzle using method of characteristics," *IJERT*, 11 2013.

[4] Charles G. Martin, "Ansys WorkBench - Fluent C-D Nozzle tutorial," 2023, accessed: 2024-11-08. [Online]. Available: https://youtu.be/P_jdywkbRQ4?si=J-zujUHOUcptdmBv